Lab 03

TITLE

Working with Git History, Stashing, and Reverting Commits

Name:

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Registration number:

(2023-BSCS-44)

Section:

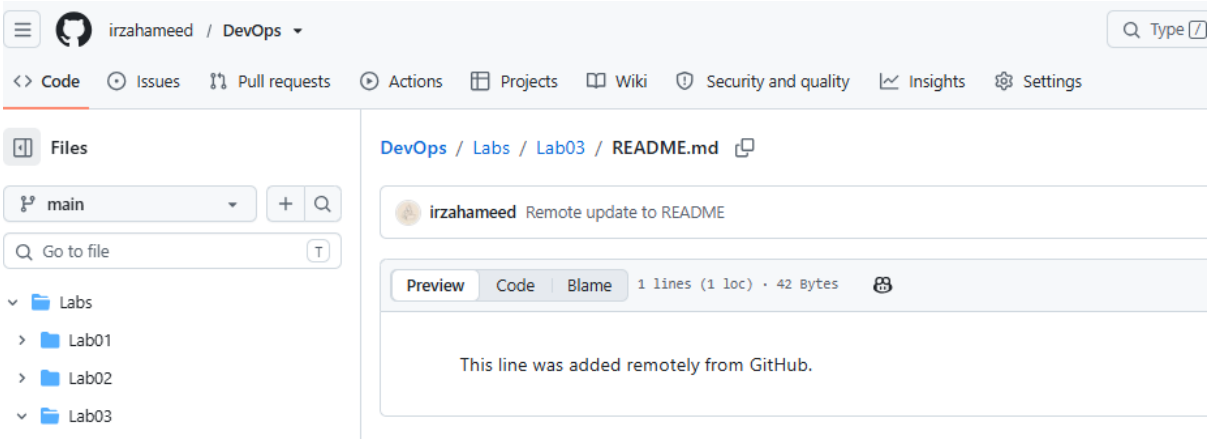
A

GitHub username:

irzahameed

Task 1 – Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase)

remote_edit.png



local_commit.png

```
irzahameed@ubunto:~/DevOps$ echo "This line was added locally.">>README.md
irzahameed@ubunto:~/DevOps$ git add README.md
irzahameed@ubunto:~/DevOps$ git commit -m "Local update to README"
[main 5862a51] Local update to README
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
irzahameed@ubunto:~/DevOps$
```

push_error.png

```
irzahameed@ubunto:~/DevOps$ git push origin main
To github.com:irzahameed/DevOps.git
 ! [rejected]        main -> main (non-fast-forward)
error: failed to push some refs to 'github.com:irzahameed/DevOps.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
irzahameed@ubunto:~/DevOps$
```

merge_commit.png

```
Merge made by the 'ort' strategy.
irzahameed@ubunto:~/DevOps$ git pull --no-rebase origin main
From github.com:irzahameed/DevOps
 * branch            main       -> FETCH_HEAD
Already up to date.
irzahameed@ubunto:~/DevOps$
```

push_after_merge.png

```
irzahameed@ubunto:~/DevOps$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
irzahameed@ubunto:~/DevOps$ _
```

rebase_pull.png

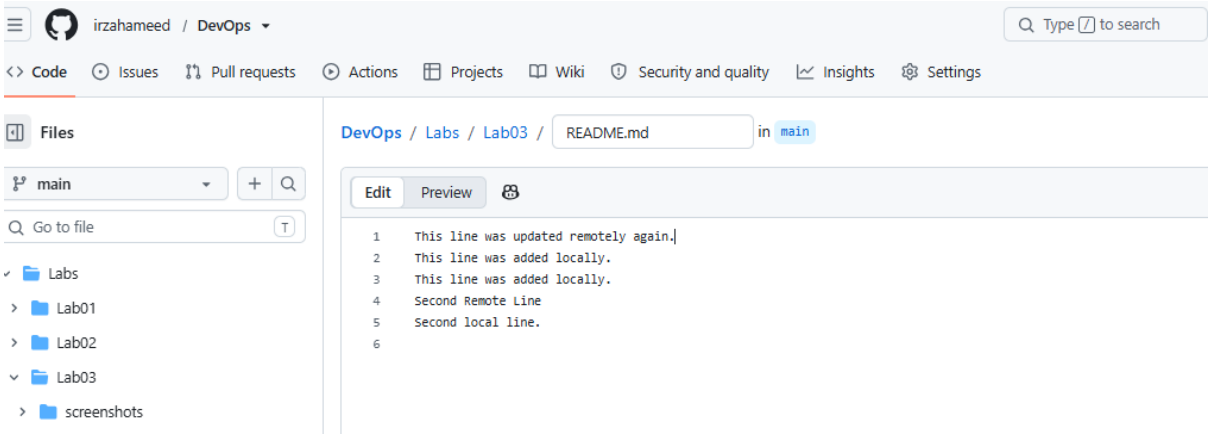
```
irzahameed@ubunto:~/DevOps$ git pull --rebase origin main
From github.com:irzahameed/DevOps
 * branch            main       -> FETCH_HEAD
Already up to date.
irzahameed@ubunto:~/DevOps$
```

push_after_rebase.png

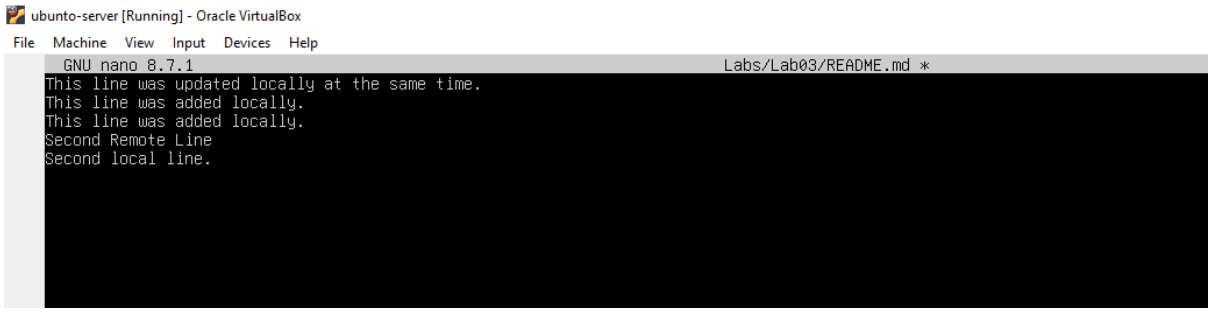
```
irzahameed@ubunto:~/DevOps$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
irzahameed@ubunto:~/DevOps$ _
```

Task 2 – Creating and Resolving Merge Conflicts Manually

remote_conflict_edit.png



local_conflict_edit.png



local_conflict_commit.png

```
irzahameed@ubunto:~/DevOps$ git add Labs/Lab03/README.md
irzahameed@ubunto:~/DevOps$ git commit -m "Local conflicting change"
[main 76ab792] Local conflicting change
 1 file changed, 1 insertion(+), 1 deletion(-)
irzahameed@ubunto:~/DevOps$ _
```

conflict_push_error.png

```
irzahameed@ubunto:~/DevOps$ git push -u origin main
To github.com:irzahameed/DevOps.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:irzahameed/DevOps.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
irzahameed@ubunto:~/DevOps$ _
```

conflict_message.png

```
irzahameed@ubunto:~/DevOps$ git pull --rebase origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.05 KiB | 119.00 KiB/s, done.
From github.com:irzahameed/DevOps
 * branch            main       -> FETCH_HEAD
   b323357..79b885f  main       -> origin/main
Auto-merging Labs/Lab03/README.md
CONFLICT (content): Merge conflict in Labs/Lab03/README.md
error: could not apply e9f0815... local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply e9f0815... # local conflicting change
irzahameed@ubunto:~/DevOps$ _
```

ubunto-server (End of lab 3 task 2) [Running] - Oracle VirtualBox

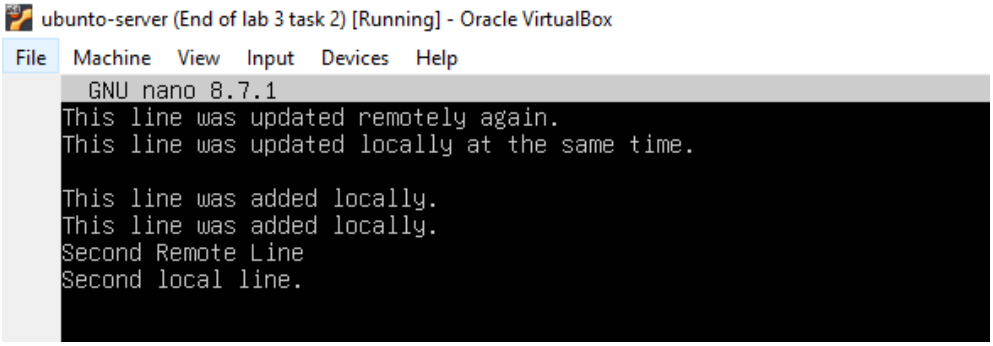
File Machine View Input Devices Help

```
  GNU nano 8.7.1                                                    L
<<<<<<< HEAD
This line was updated remotely again.
This line was updated locally at the same time.

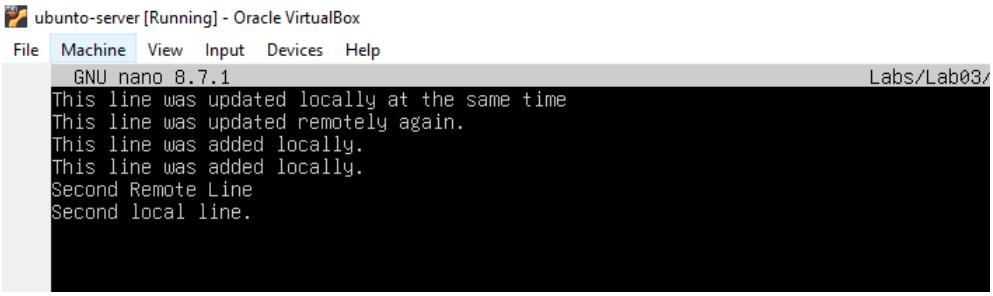
=======

This line was updated locally at the same time
>>>>>>> e9f0815 (local conflicting change)

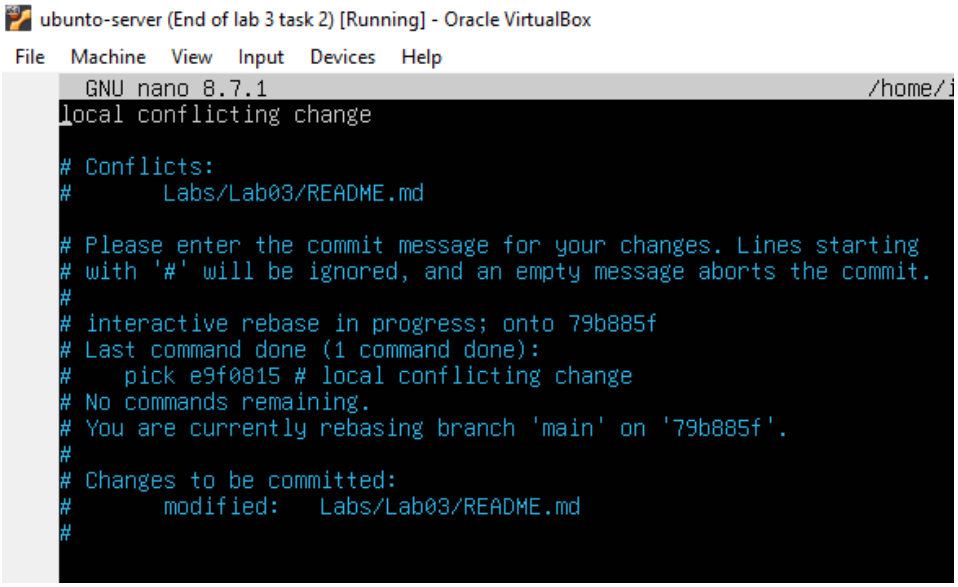
This line was added locally.
This line was added locally.
Second Remote Line
Second local line.
```



resolved_readme.png



rebase_continue.png



push_after_resolve.png

```
Successfully rebased and updated refs/heads/main.
irzahameed@ubunto:~/DevOps$ git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 435 bytes | 19.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To github.com:irzahameed/DevOps.git
   79b885f..3dd9770  main -> main
branch 'main' set up to track 'origin/main'.
irzahameed@ubunto:~/DevOps$
```

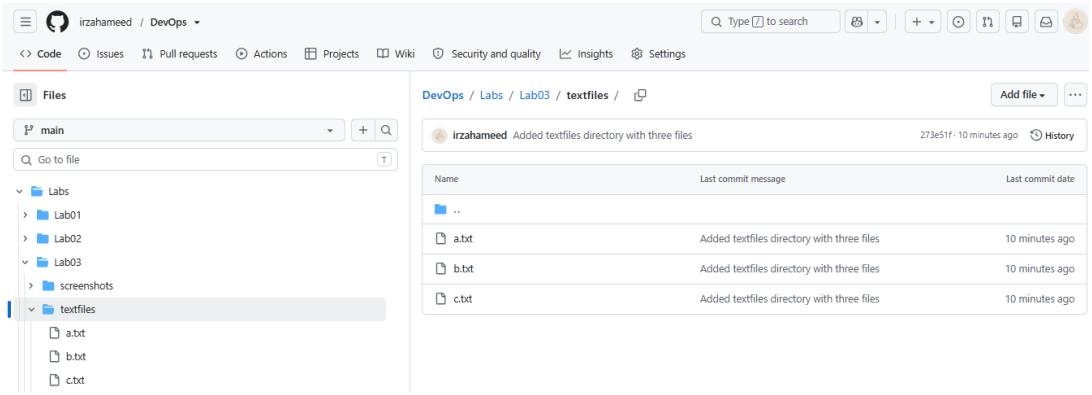
push_textfiles.png

```
irzahameed@ubunto:~/DevOps$ mkdir -p Labs/Lab03/textfiles
irzahameed@ubunto:~/DevOps$ echo "File A content" > Labs/Lab03/textfiles/a.txt
irzahameed@ubunto:~/DevOps$ echo "File B content" > Labs/Lab03/textfiles/b.txt
irzahameed@ubunto:~/DevOps$ echo "File C content" > Labs/Lab03/textfiles/c.txt
irzahameed@ubunto:~/DevOps$ git add .
irzahameed@ubunto:~/DevOps$ git commit -m "Added textfiles directory with three files "
[main 273e51f] Added textfiles directory with three files
 3 files changed, 3 insertions(+)
 create mode 100644 Labs/Lab03/textfiles/a.txt
 create mode 100644 Labs/Lab03/textfiles/b.txt
 create mode 100644 Labs/Lab03/textfiles/c.txt
irzahameed@ubunto:~/DevOps$ git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Compressing objects: 100% (5/5), done.
Writing objects: 100% (8/8), 584 bytes | 58.00 KiB/s, done.
Total 8 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:irzahameed/DevOps.git
   3dd9770..273e51f  main -> main
irzahameed@ubunto:~/DevOps$
```

gitignore_push.png

```
irzahameed@ubunto:~/DevOps$ echo "textfiles/*" > Labs/Lab03/ .gitignore
-bash: Labs/Lab03/: Is a directory
irzahameed@ubunto:~/DevOps$ git add Labs/Lab03/ .gitignore
fatal: pathspec '.gitignore' did not match any files
irzahameed@ubunto:~/DevOps$ git add Labs/Lab03/ .gitignore
fatal: pathspec '.gitignore' did not match any files
irzahameed@ubunto:~/DevOps$ echo "textfiles/*" > Labs/Lab03/.gitignore
irzahameed@ubunto:~/DevOps$ cat Labs/Lab03/.gitignore
textfiles/*
irzahameed@ubunto:~/DevOps$ git add Labs/Lab03/.gitignore
irzahameed@ubunto:~/DevOps$ git push origin main
Everything up-to-date
irzahameed@ubunto:~/DevOps$ _
```

repo_still_has_textfiles.png

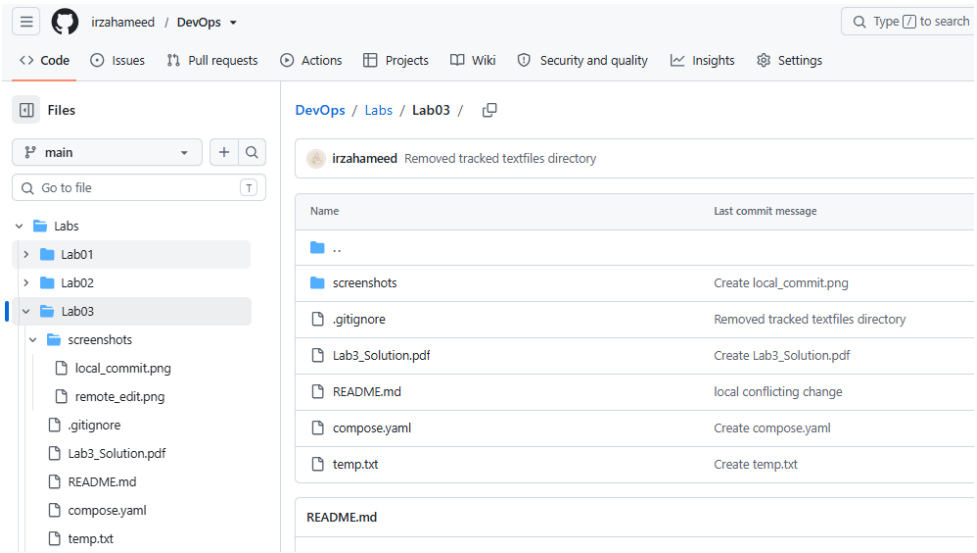


rm_cached_push.png

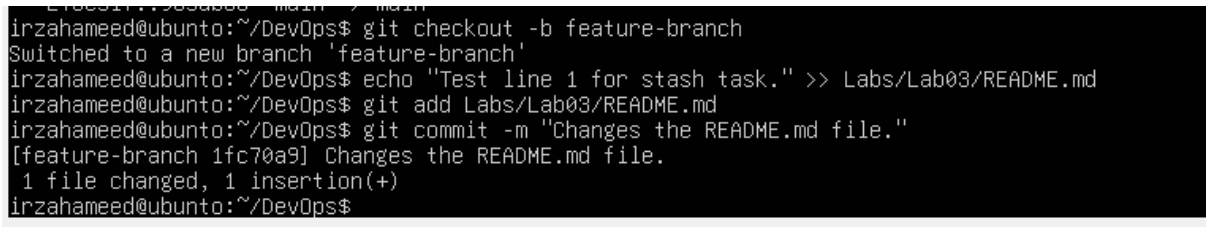
```
irzahameed@ubunto:~/DevOps$ git rm -r --cached Labs/Lab03/textfiles
rm 'Labs/Lab03/textfiles/a.txt'
rm 'Labs/Lab03/textfiles/b.txt'
rm 'Labs/Lab03/textfiles/c.txt'
irzahameed@ubunto:~/DevOps$ git add.
git: 'add.' is not a git command. See 'git --help'.

The most similar command is
        add
irzahameed@ubunto:~/DevOps$ git add .
irzahameed@ubunto:~/DevOps$ git commit -m "Removed tracked textfiles directory"
[main 965ab68] Removed tracked textfiles directory
 4 files changed, 1 insertion(+), 3 deletions(-)
 create mode 100644 Labs/Lab03/.gitignore
 delete mode 100644 Labs/Lab03/textfiles/a.txt
 delete mode 100644 Labs/Lab03/textfiles/b.txt
 delete mode 100644 Labs/Lab03/textfiles/c.txt
irzahameed@ubunto:~/DevOps$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 436 bytes | 27.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:irzahameed/DevOps.git
   273e51f..965ab68  main -> main
irzahameed@ubunto:~/DevOps$
```

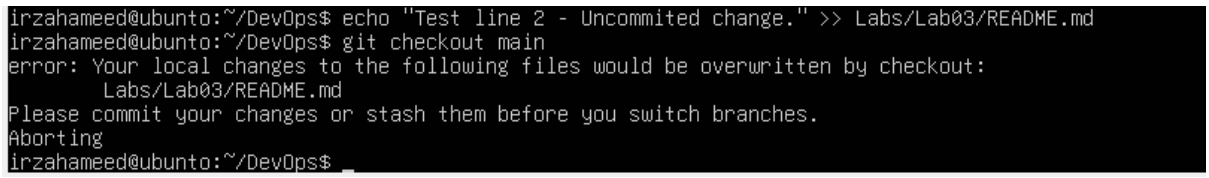
repo_textfiles_removed.png



modified_readme.png



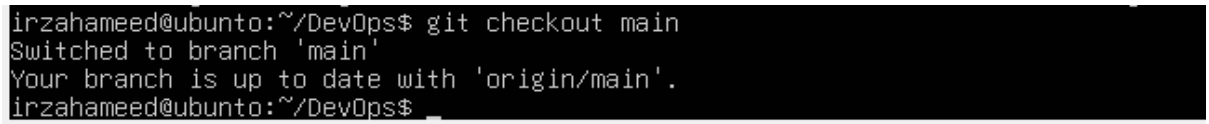
checkout_error.png



stash_command.png



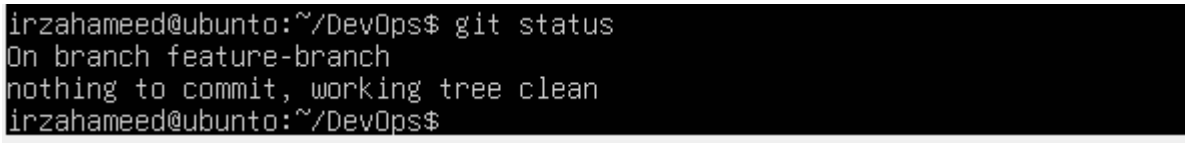
branch_switched.png



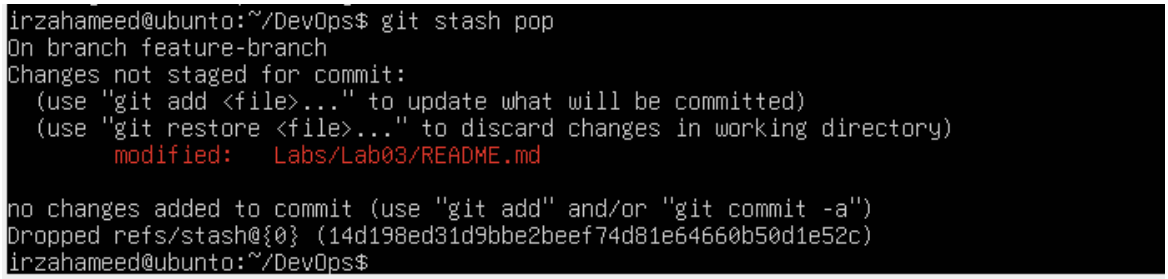
back_to_feature.png



status_clean.png

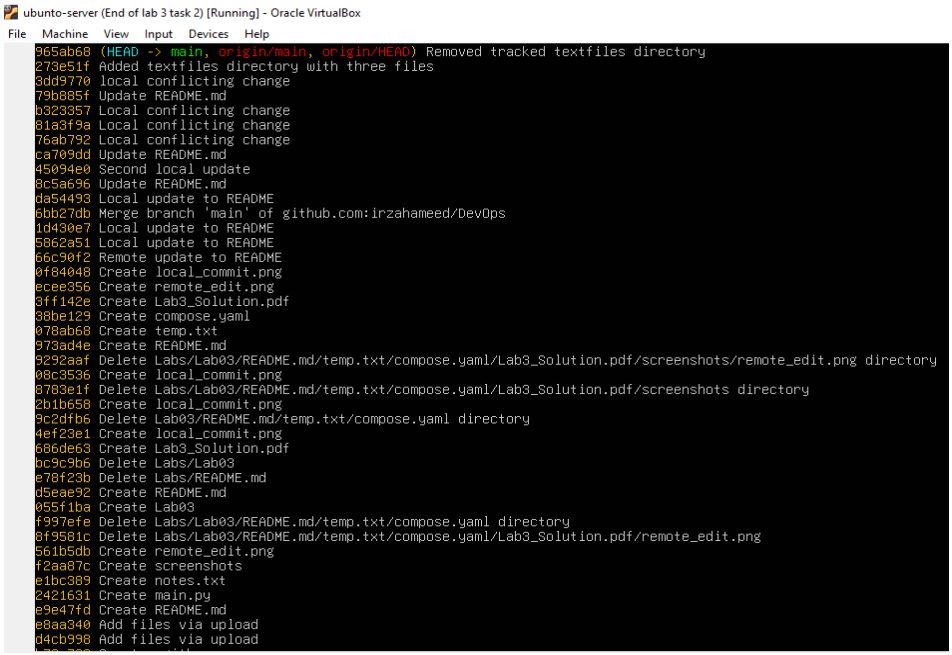


stash_pop.png

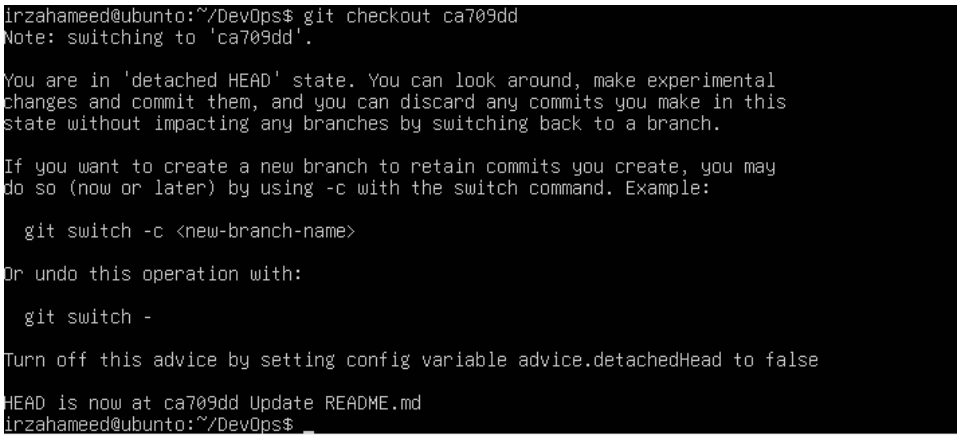


Task 5 – Checkout a Specific Commit Using git log

log_before_checkout.png



detached_head.png



back_to_main.png



Task 6 – Resetting Commits (Soft vs Hard Reset) (With Verification Steps)

first_commit.png

```
irzahameed@ubunto:~/DevOps$ echo "This is the first test line for reset." >> Labs/Lab03/README.md
irzahameed@ubunto:~/DevOps$ git add .
irzahameed@ubunto:~/DevOps$ git commit -m "Added test line."
[main ac3d560] Added test line.
 1 file changed, 1 insertion(+)
irzahameed@ubunto:~/DevOps$
```

second_commit.png

```
irzahameed@ubunto:~/DevOps$ echo "This is the Second test line for reset." >> Labs/Lab03/README.md
irzahameed@ubunto:~/DevOps$ git add .
irzahameed@ubunto:~/DevOps$ git commit -m "Second test commit"
[main 4306f8e] Second test commit
 1 file changed, 1 insertion(+)
irzahameed@ubunto:~/DevOps$
```

log_before_reset.png

```
irzahameed@ubunto:~/DevOps$ git log --oneline -3
4306f8e (HEAD -> main) Second test commit
ac3d560 Added test line.
965ab68 (origin/main, origin/HEAD) Removed tracked textfiles directory
irzahameed@ubunto:~/DevOps$ _
```

file_before_reset.png

```
irzahameed@ubunto:~/DevOps$ cat Labs/Lab03/README.md
This line was updated remotely again.
This line was updated locally at the same time.

This line was added locally.
This line was added locally.
Second Remote Line
Second local line.
This is the first test line for reset.
This is the Second test line for reset.
irzahameed@ubunto:~/DevOps$ _
```

soft_reset.png

```
irzahameed@ubunto:~/DevOps$ git reset --soft HEAD~1
irzahameed@ubunto:~/DevOps$ _
```

log_after_soft_reset.png

```
irzahameed@ubunto:~/DevOps$ git reset --soft HEAD~1
irzahameed@ubunto:~/DevOps$ git log --oneline -3
ac3d560 (HEAD -> main) Added test line.
965ab68 (origin/main, origin/HEAD) Removed tracked textfiles directory
273e51f Added textfiles directory with three files
irzahameed@ubunto:~/DevOps$
```

file_after_soft_reset.png

```
irzahameed@ubunto:~/DevOps$ cat Labs/Lab03/README.md
This line was updated remotely again.
This line was updated locally at the same time.

This line was added locally.
This line was added locally.
Second Remote Line
Second local line.
This is the first test line for reset.
This is the Second test line for reset.
irzahameed@ubunto:~/DevOps$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   Labs/Lab03/README.md

irzahameed@ubunto:~/DevOps$
```

file_after_hard_reset.png

```
irzahameed@ubunto:~/DevOps$ git commit -m "Second test commit"
[main 74b9463] Second test commit
 1 file changed, 1 insertion(+)
irzahameed@ubunto:~/DevOps$
```

hard_reset.png

```
irzahameed@ubunto:~/DevOps$ git reset --hard HEAD~1
HEAD is now at ac3d560 Added test line.
irzahameed@ubunto:~/DevOps$
```

log_after_hard_reset.png

```
irzahameed@ubunto:~/DevOps$ git log --oneline -3
ac3d560 (HEAD -> main) Added test line.
965ab68 (origin/main, origin/HEAD) Removed tracked textfiles directory
273e51f Added textfiles directory with three files
irzahameed@ubunto:~/DevOps$
```

file_after_hard_reset.png

```
irzahameed@ubunto:~/DevOps$ cat Labs/Lab03/README.md
This line was updated remotely again.
This line was updated locally at the same time.

This line was added locally.
This line was added locally.
Second Remote Line
Second local line.
This is the first test line for reset.
irzahameed@ubunto:~/DevOps$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
irzahameed@ubunto:~/DevOps$
```

Task 7 – Amending the Last Commit

first_amend_commit.png

```
irzahameed@ubunto:~/DevOps$ echo "This is an incomplete commit" >> Labs/Lab03/temp.txt
irzahameed@ubunto:~/DevOps$ git add .
irzahameed@ubunto:~/DevOps$ git commit -m "Fix log message"
[main d4a0b73] Fix log message
 1 file changed, 1 insertion(+)
irzahameed@ubunto:~/DevOps$
```

amend_commit.png

```
[main 92c4645] Fix log message
 Date: Tue Oct 20 10:04:20 2026 +0000
 2 files changed, 2 insertions(+)
irzahameed@ubunto:~/DevOps$
```

Task 8 – Reverting a Commit (Safe Undo on Remote Branch)

commit_temp_file.png

```
irzahameed@ubunto:~/DevOps$ echo "temporary text" >> Labs/Lab03/temp.txt
irzahameed@ubunto:~/DevOps$ git add .
irzahameed@ubunto:~/DevOps$ git commit -m "Added temporary text file"
[main 9187c3d] Added temporary text file
 1 file changed, 1 insertion(+)
irzahameed@ubunto:~/DevOps$ git push origin main
```

revert_commit.png

```
irzahameed@ubunto:~/DevOps$ git log --oneline -3
9187c3d (HEAD -> main, origin/main, origin/HEAD) Added temporary text file
ac3d560 Added test line.
965ab68 Removed tracked textfiles directory
```

ubunto-server (End of lab 3 task 2) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
  GNU nano 8.7.1                                                  /home/
Revert "Added temporary text file"

This reverts commit 9187c3df69982268d16076e66f0a42a0d6bad610.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch main
# Your branch is up to date with 'origin/main'.
#
# Changes to be committed:
#       modified:   Labs/Lab03/temp.txt
#
```

```
[main 3afb5fb] Revert "Added temporary text file"
 1 file changed, 1 deletion(-)
irzahameed@ubunto:~/DevOps$
```

revert_push.png

```
irzahameed@ubunto:~/DevOps$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 416 bytes | 46.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:irzahameed/DevOps.git
   9187c3d..3afb5fb  main -> main
irzahameed@ubunto:~/DevOps$
```

Task 9 – Force Push (With Caution)

new_branch.png

```
irzahameed@ubunto:~/DevOps$ git checkout -b test-force
Switched to a new branch 'test-force'
irzahameed@ubunto:~/DevOps$
```

force_commit.png

```
irzahameed@ubunto:~/DevOps$ echo "Force push test" >> Labs/Lab03/temp.txt
irzahameed@ubunto:~/DevOps$ git add .
irzahameed@ubunto:~/DevOps$ git commit -m "Force push test commit"
[test-force 2ae19dc] Force push test commit
 1 file changed, 1 insertion(+)
irzahameed@ubunto:~/DevOps$
```

push_force_branch.png

```
irzahameed@ubunto:~/DevOps$ git push origin test-force
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 415 bytes | 83.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/irzahameed/DevOps/pull/new/test-force
remote:
To github.com:irzahameed/DevOps.git
 * [new branch]      test-force -> test-force
irzahameed@ubunto:~/DevOps$
```

hard_reset_force.png

```
irzahameed@ubunto:~/DevOps$ git reset --hard HEAD~1
HEAD is now at 3afb5fb Revert "Added temporary text file"
irzahameed@ubunto:~/DevOps$
```

normal_push.png

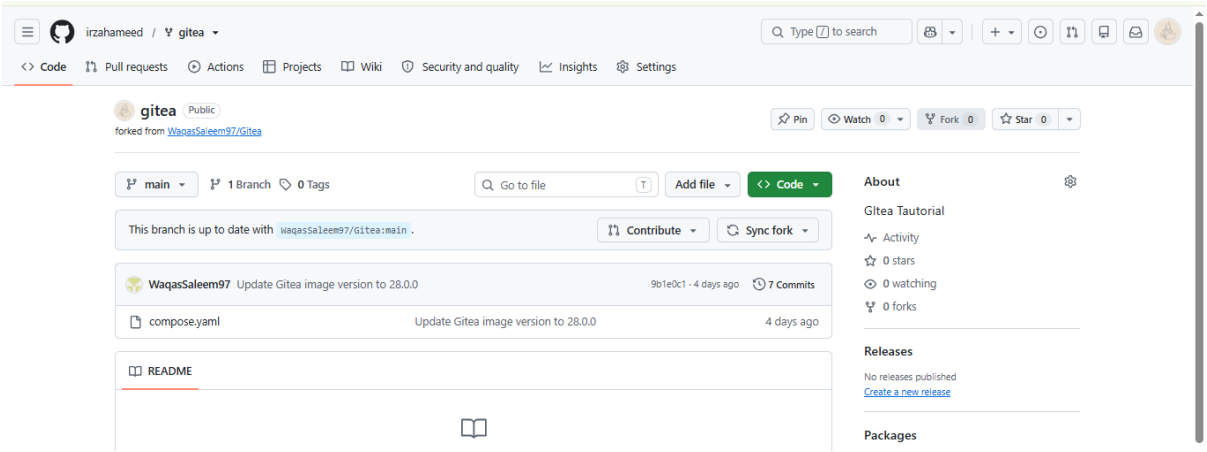
```
irzahameed@ubunto:~/DevOps$ git push origin test-force
To github.com:irzahameed/DevOps.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:irzahameed/DevOps.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
irzahameed@ubunto:~/DevOps$
```

force_push.png

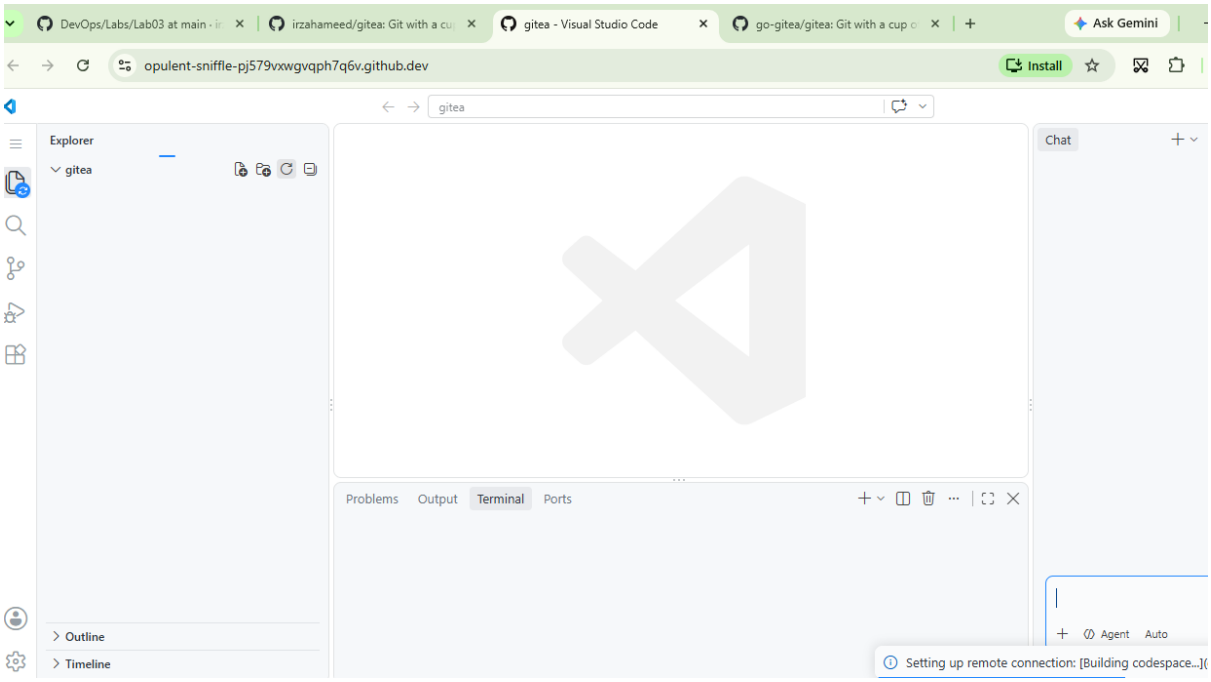
```
irzahameed@ubunto:~/DevOps$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:irzahameed/DevOps.git
 + 2ae19dc...3afb5fb test-force -> test-force (forced update)
irzahameed@ubunto:~/DevOps$
```

Task 10 – Running Gitea in GitHub Codespaces via Docker Compose

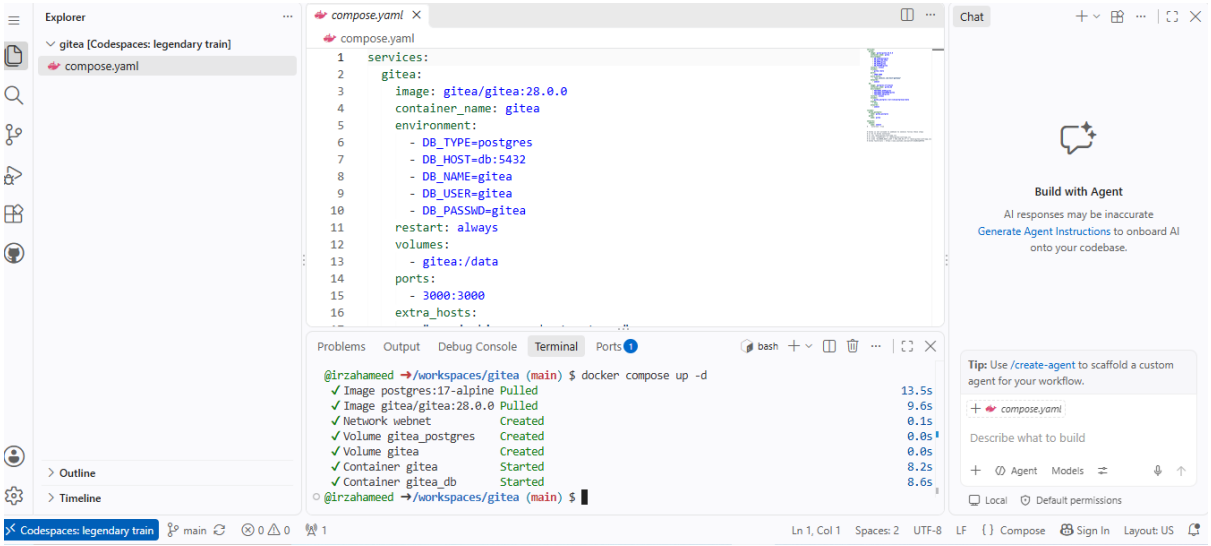
forked_gitea.png



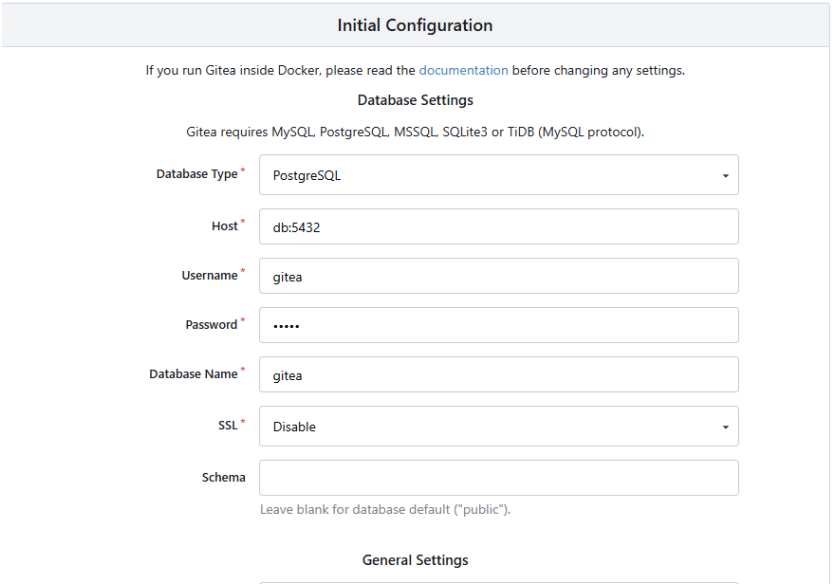
codespace_loading.png



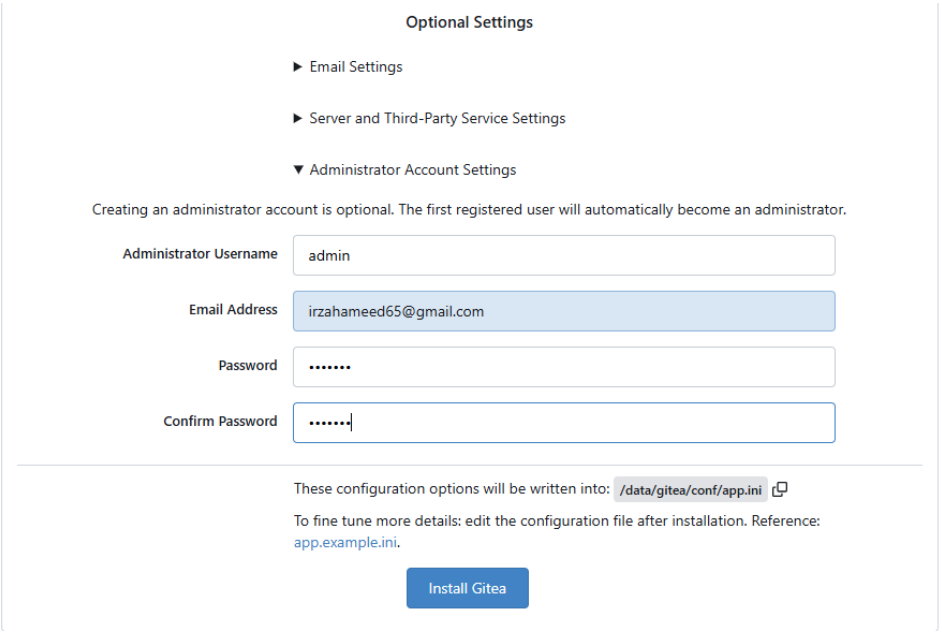
docker_up.png



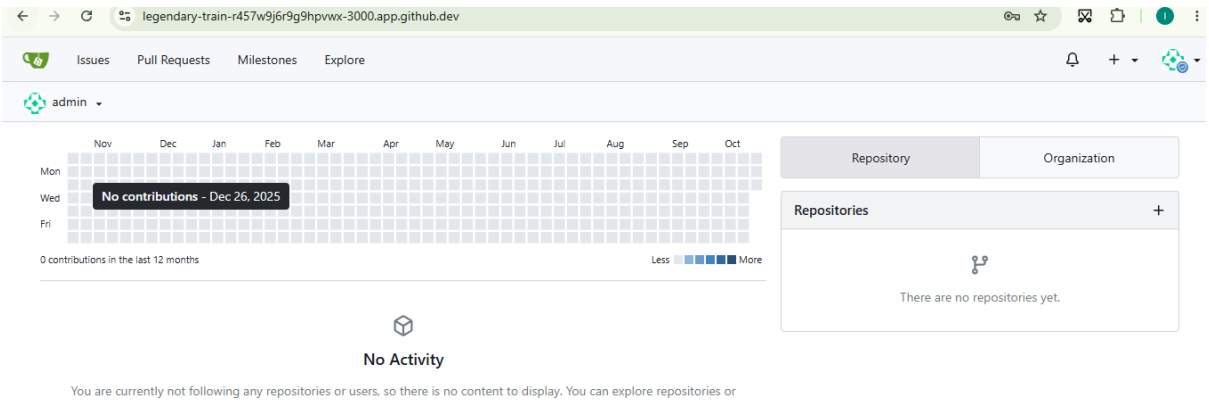
gitea_install_page.png



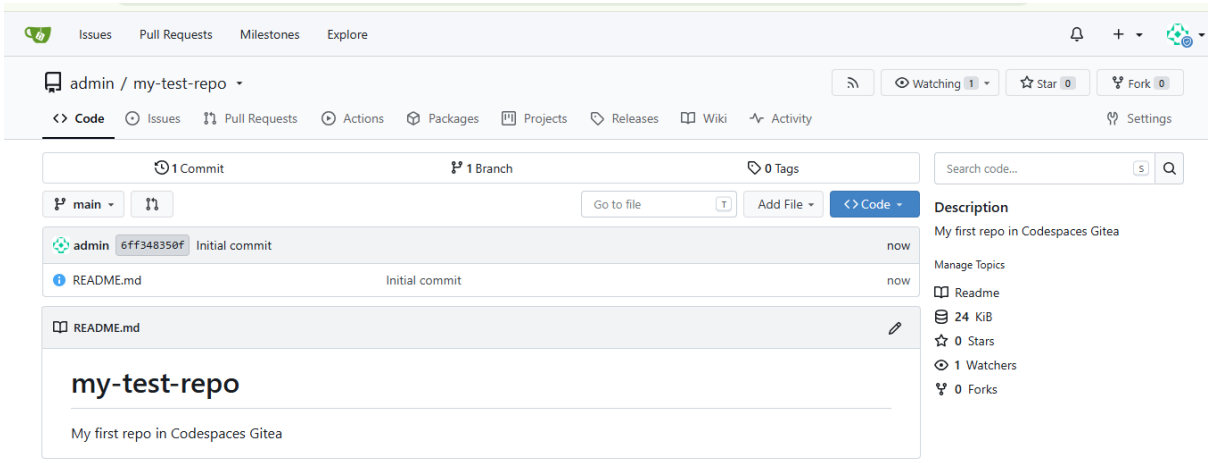
admin_setup.png



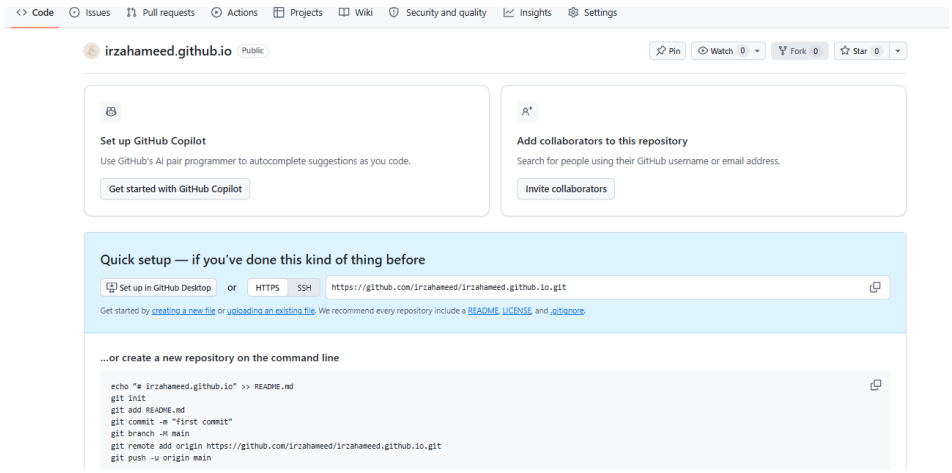
gitea_dashboard.png



gitea_new_repo.png



github_pages_repo.png



github_pages_settings.png

